

TYPE I & II ERRORS

**Module 2: The Shape of Chaos **

1. THE INEVITABILITY OF MISTAKES

In statistical testing, your computer is looking at chaotic, overlapping Bell Curves and trying to make a hard “Yes or No” decision. Because data is fundamentally probabilistic, **you will eventually make a mistake.**

Statisticians categorize these mistakes into two distinct, unavoidable traps known as **Type I** and **Type II Errors**.

2. ANATOMY OF THE CONFUSION MATRIX

To map every possible outcome of a decision, we use a 2x2 grid called a **Confusion Matrix**. There are two ways to be right, and two ways to be wrong:

- **True Positive (Hit):** Reality says something happened, and your math correctly detected it. *(The patient is sick, and the test came back positive).*
- **True Negative (Correct Rejection):** Reality says nothing happened, and your math correctly ignored it. *(The patient is healthy, and the test came back negative).*

✗ TYPE I ERROR (FALSE POSITIVE | α)

- **The Definition:** You mathematically declare a discovery, but reality was actually just pure coincidence. You **rejected a true Null Hypothesis** (H_0).
- **The Analogy:** A smoke detector going off when you are just boiling water. A jury sending an innocent citizen to prison.
- **The Math:** The probability of committing a Type I error is exactly equal to your chosen Significance Level (α). If you set $\alpha = 0.05$, you are accepting a 5% baseline risk of crying wolf.

✗ TYPE II ERROR (FALSE NEGATIVE | β)

- **The Definition:** Reality genuinely changed, but your math was too blunt or your sample size was too small to notice it. You **failed to reject a false Null Hypothesis** (H_0).
- **The Analogy:** A smoke detector staying totally silent while the kitchen is on fire. A jury letting a guilty criminal walk free.
- **The Math:** The probability of committing a Type II error is represented by the Greek letter Beta (β).

3. THE ULTIMATE STATISTICAL PARADOX

Why can't we just engineer a test that has a 0% chance of False Positives and a 0% chance of False Negatives?

Because α and β **sit on a sliding see-saw**. They are mathematically bound to one another across overlapping probability curves.

- If you make your Significance threshold extremely strict ($\alpha = 0.001$) to make sure you *never* convict an innocent person (Type I), you make your scalpel so blunt that you fail to detect genuine discoveries, causing your False Negatives (β) to skyrocket.
- If you make your threshold extremely sensitive so you catch every possible disease (Type II), your False Positives (α) explode.

You cannot eliminate one statistical error without increasing the other.

4. STATISTICAL POWER ($1 - \beta$)

If β is the probability of missing a real discovery, then the exact opposite—**the probability of correctly detecting a true discovery**—is known as Statistical Power.

$$\text{Power} = 1 - \beta$$

- Think of Power as the “resolution” or “lens quality” of your mathematical microscope.
- **How to crank up your Power:** The absolute best way to increase your Power without ruining your False Positive rate (α) is to **drastically increase your sample size (N)**. A bigger sample size pulls the overlapping Bell Curves apart, giving your math a crystal clear view of reality.